

MetropoLive

Group 17 - Sunsets

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<https://github.com/CMPT-276-SUMMER-2025/final-project-17-sunsets>

Overview:

The application will be able to find events nearby along with relevant information such as transit to the event and perhaps a calendar.

The idea for this application came from people needing a way to meet new people and go outside more. With the lack of third places nowadays, interacting with people and making connections is harder than ever. People are lonelier than ever, making them depressed. This application assists in resolving those problems. The application will also save time for those busy people who can't afford to spend time searching for events themselves. People foreign to the area will benefit from being able to easily access local events.

Potential users:

- University students looking to socialize, but have little time to search and plan due to busy schedules
- Tourists looking for events, but are unfamiliar with local transit
- Volunteers looking for opportunities

Persona for each application group:

Liam:

- Computer science university student
- Limited time because of busy schedule
- Has a hard time talking to fellow students

Goals:

- Meet like minded people
- Make new friends
- Get out of the house more
- Save money

Sofia:

- Tourist from America
- Spontaneous
- Loves to meet new people

Goals:

- Experience the local culture
- Try new foods
- Meet new people
- Have fun

Charlie:

- High school student
- Looking for volunteering opportunities before applying to college
- Doesn't own a car

Goals:

- Gain volunteer hours
- Get hands-on experience
- Meet people and network

List of chosen APIs (4 in total):

Primary:

Google Maps Platform Routes API: The official API for the familiar Google Maps interface. Like MapBox Directions, this API is responsible for calculating optimal routes between two or more locations for several modes of transportation, while also taking into account current road conditions.

Eventbrite API: The official API for Eventbrite. Eventbrite allows users to find events based on location, price, and topic. Events such as concerts, festivals, meetups, workshops, etc.

(Optional) Add to calendar button: Allows connection to calendar apps to keep track of signed up events. Instead of using another API, this feature can be implemented by generating a universal calendar file (extension “.ics”) directly in the user's browser with JavaScript. A library that may be useful for this button could be “ics.js”.

Secondary:

MapBox Directions API: The official API for MapBox, a provider for custom online maps. Calculates ideal routes for multiple modes of transportation, accounting for real-time traffic conditions.

Meetup API: The official API of Meetup. Meetup provides users with information on local crowd-sourced events.

Description of features for each API & User stories:

Primary:

Google Maps Platform Routes API:

- Shows optimal route between two locations
 - *“I want to see the fastest route to the event so I can get there on time for my volunteer work.”*
- Shows turn-by-turn instructions
 - *“I’m not familiar with the local transit, so I want to know which buses I need to take and what stops to get off at.”*
- Swap between transit profiles

- *"I don't own a car, so I want to be able to see directions for either walking or cycling."*

Eventbrite API:

- Can search for nearby events
 - *"I want to discover new events near me so I can meet new people."*
- Can search by category
 - *"I like listening to live music, so I want to search for concerts."*
- Shows information about events such as price
 - *"I want to be able to see details about events so I can see what matches with my budget and schedule."*

Secondary:

MapBox Directions API:

- Displays optimized routes between two or more locations
 - *"I want to see the fastest routes between a handful of events that I want to attend."*
- Shows turn-by-turn instructions
 - *"The transit options near me are unfamiliar, so I want to know which buses or walking routes I need to take and what stops to make along the way."*
- Quickly switch between different modes of transit
 - *"I don't always have easy access to a car, so I want to be able to see directions for either walking, cycling, or occasionally driving."*

Meetup API:

- Search for events by keywords and filters
 - *"I want to be able to search for specific events that fit my interests."*
- Shows photos associated with events
 - *"I want to see photos of events so I can get a sense of what to expect before I go."*
- Provides description of events
 - *"I want a detailed description of events so I can choose one I can afford."*

Add to calendar button:

- Can connect to any major calendar app like Outlook or Apple
 - *"I need to manage all of my events effectively, so incorporating them into one app would be best."*
- Can make edits in the calendar app
 - *"I want to be able to modify or erase information quickly given that a problem occurs or the timings change."*
- Provides easy access between app and calendar
 - *"I would like for there to be a way to access the app with ease through the calendar so that I do not have to manually switch between them."*

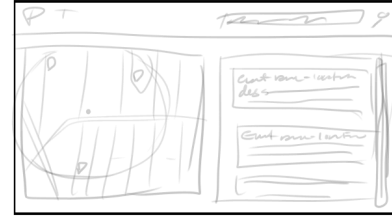
Low-fidelity storyboard of the application interface/features:



◦ First enter your location



◦ Map shows location
◦ Select preferences then search



◦ Shows locations of events
◦ Shows list of events

◦ Click on an event to see info (Name, location, Price, category, time, date, etc.)

◦ Scroll down to find route to event

◦ Select between transport types and see directions



Chosen front-end technology stack:

- React JS
- CSS modules
- HTML

Explanation why we chose this stack:

- We can code a combination of HTML and javascript together using JSX which makes it easier to visualize and understand the code.
- We can reuse components for instance, we could reuse event cards.
- Uses a virtual DOM so updates are made faster as it only renders the parts of the web page that we change and not the whole web page.
- We will use CSS modules to prevent stylings across the entire web page, this will help us to style specific components or tags within those components.